SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID – 26166**
- **Problem Statement Title – Multi-modal, Sun angle and scale invariant image correspondence using Chandrayaan-2 optical images (OHRC, TMC and IIRS)**
- **Theme – Space Technology**
- **PS Category – Software**
- **Team ID – To be added once allotted on the SIH portal**
- **Team Name (Registered on portal) – Shevolution**

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IDEA TITLE

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❖Proposed Solution (Describe your Idea/Solution/Prototype)

- **Detailed explanation:** geo.align_pair scale-normalizes CH2 OHRC vs LRO NAC onto one grid; SIFT, a custom mod-π illumination-invariant descriptor, and LightGlue are then benchmarked on the same ground truth.
- **How it addresses the problem:** sun-angle change collapses SIFT (inliers ~1580 → single digits by 30°); LightGlue holds 80–94% grid coverage through 60° + of sun-angle difference where classical descriptors fail.
- **Innovation:** fit_reliability auto-flags registrations whose inliers reduce to ≤4 unique points as untrustworthy; results export natively as an ISIS PVL control network for ISRO's jigsaw pipeline.

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TECHNICAL APPROACH

- **Technologies:** Python, OpenCV, NumPy/SciPy, Shapely, PyTorch (LightGlue + SuperPoint), rasterio, pdr (PDS3/4), webgeocalc (NAIF SPICE), pvl (ISIS control-net); 188 automated pytest cases.
- **Methodology:** Product (CH2 OHRC | LRO NAC) → align_pair (common-grid resample) → prep (contrast-norm, mod-π orientation) → match/match_tiled (SIFT | mod-π | LightGlue, MAGSAC, sub-pixel) → metrics (RMSE, coverage, fit_reliability) → deliverable + ISIS control network.

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FEASIBILITY AND VIABILITY

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- **Feasibility:** 188/188 automated tests pass on hand-checkable synthetic ground truth; real PDS3/PDS4 parsing and NAIF SPICE geometry confirmed end-to-end on genuine CH2/LRO data, all open-source.
- **Challenges and risks:** on a real CH2 × LRO pair (~26.5% overlap), align_pair's single 4-corner perspective fit drifts along a long, narrow, curved-orbit strip — no trustworthy registration there yet.
- **Strategy:** a multi-control-point / piecewise geometric model for align_pair is the scoped next step; fit_reliability surfaces this limitation honestly instead of a false win.

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IMPACT AND BENEFITS

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- **Potential impact on the target audience:** cuts manual tie-point selection effort in ISRO's photogrammetry pipeline for sun-angle-mismatched pairs otherwise discarded as unusable.
- **Benefits:** cross-mission data fusion (Chandrayaan-2 × LRO) despite differing instruments/illumination; self-verified output means no silent false positives; extensible to any georeferenced planetary matching problem.

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RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2026

- Details / Links of the reference and research work
- mod-π's basis: RIFT, HAPCG illumination-robust remote-sensing matching literature.
- LightGlue / SuperPoint — strongest benchmarked matcher in this project's own testing.
- USGS ISIS control-network (PVL) format — validated via the independent pvl parser.
- PDS3 (LRO) / PDS4 (Chandrayaan-2) label specs — source formats read natively.
- NAIF SPICE / WebGeocalc — real solar-geometry (incidence, subsolar azimuth) for LRO.